

# HoloInset: 3D Biomedical Image Data Exploration through Augmented Hologram Insets

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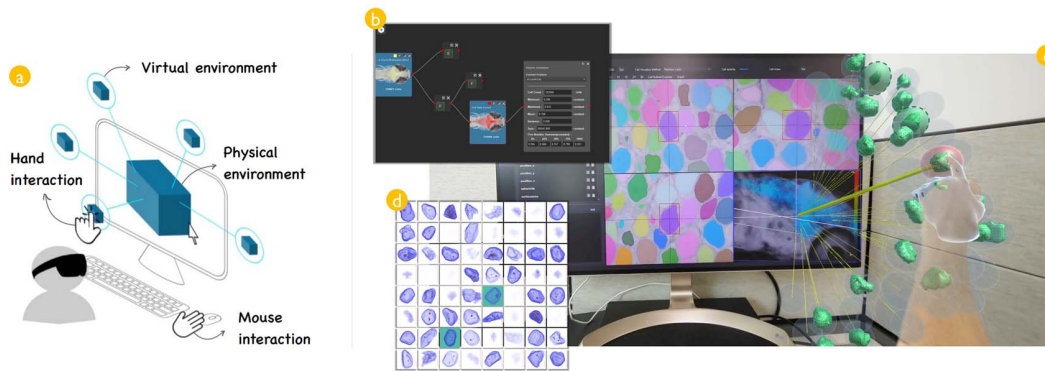


Figure 1: Concept of *HoloInset* framework — (a): The framework can extend conventional visual analytics environment to mixed reality. ZeVis [1] with *HoloInset*. — (b): The inset generation event occurs when the representative cell generation function is executed in the graph-based data processing interface of the physical environment. (c): Left-hand *Focus* interaction (red sphere) in virtual environment shows the detailed cell information in the physical environment. (d): Left-hand *Select* interaction (green sphere) in virtual environment adds the cells into the selected list in the physical environment (green background).

## ABSTRACT

The extended reality (XR) provides realistic depth perception and huge visualization spaces, which can serve as a powerful workspace for 3D data exploration and analysis. However, a direct adaptation of XR to conventional 3D data exploration tasks is less feasible due to several hardware limitations, such as low screen resolution, dizziness, narrow field of view, etc.

In this paper, we propose a novel mixed reality visualization scheme, *HoloInset*, which combines a conventional visual analytics system and a virtual environment to effectively explore 3D biomedical image data. We also demonstrate the usability of the proposed visualization through a real-world analysis case.

**Index Terms:** Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality

## 1 INTRODUCTION

3D biomedical images are complex with the characteristics of large and three-dimensional (3D) volumetric. Most conventional studies to analyze these data were mostly based on a two-dimensional (2D) desktop monitor. The desktop monitor suffers from the limitation in usability when dealing with large-scale 3D visualization, such as bothersome of switching back and forth between multiple visualizations or the limited working space confined to a physical 2D screen.

Extended reality (XR) is characterized to provide a 3D visualization that allows users to feel the realistic 3D depth perception and to use all the surrounding space as a visualization space. These

characteristics can be applied as a great advantage when visualizing and analyzing 3D data. Therefore, immersive analytics, which is a research field to study the use of XR for data analysis, is emerging for 3D data analysis [2, 5]. However, several studies reported that XR devices might be inadequate for some analysis tasks due to hardware limitations [4, 5]. Such hardware limitations make the 3D analysis task difficult, especially when dealing with large, complex 3D biomedical images that require detailed work.

In this study, we propose a new mixed reality visualization method, *HoloInset* (Fig. 1a). The proposed method can be seamlessly combined a virtual analysis environment with conventional desktop-based visual analytics systems without disrupting the existing visual analytics workflow while providing additional assistance through augmented hologram insets. We also demonstrate the proposed method with a real-analysis case.

## 2 HOLOINSET

*HoloInset* is based on the inset visualization paradigm [3] which shows several sub-visualizations around the main visualization for highlighting important information and specific parts of the main visualization. This method can provide additional information over the main visualization without impairing visibility. Our proposed method assumes the physical environment as the main visualization and represents virtual visualizations as insets, which connects the two environments effectively.

The hologram insets are created based on the information received from the 2D physical environment and can provide depth perception of 3D data to users. In addition, it plays the role of communicator between the 2D physical and 3D virtual environments. The hologram inset consists of three components (i.e., 3D data visualization, interactable holder, and leader line) (Fig. 2), and is placed in a 3D virtual space defined by the user's gaze and the workspace of the physical screen where the existing analysis work is performed.

In the following, we explain how *HoloInset* connects two environments using a real-world analysis scenario (Fig. 1b-d).

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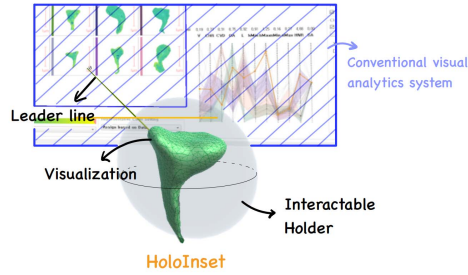


Figure 2: Visualization components of a hologram inset. The 3D surface mesh of the dendritic spine surrounded by an interactable holder is placed in a virtual environment, and the leader line points to the widget of the physical environment connected to this inset.

## 2.1 ZeVis with *HoloInset*: Large-scale data exploration

ZeVis [1] is a visual analytics system developed to explore the numerous cell objects in terabyte-scale whole-brain larval zebrafish microscopy data. We combined *HoloInset* with one function of this system, *Representative cell exploration* (Fig. 1d). The function creates 64 clusters to inspect the morphology of individual cells (over 200,000 different cells) and shows the representative cell shape of each cluster grouped by the shape similarity.

In the existing system, in order to analyze cell location distribution or select target cells (e.g., normal cell, or merged cell), users should observe the volume rendering of each cell in the Representative cell explorer (Fig. 1d) to check the approximate shape of the cell, and then click the cell to change the view into the Visualization view (the monitor of Fig. 1c) to confirm the detailed cell shape and location. There are two inconveniences here. First, users cannot see both views at once due to the visualization space limitation, so users have to click all the cells in the Representative cell explorer to check the detailed information at the Visualization view. Second, since depth information is lost in each cell visualization, it is necessary to rotate the cell several times to grasp the 3D shape. These two inconveniences can be addressed through *HoloInset* framework.

**Inset generation event:** When the user activates the *Representative cell generation function* in the Graph-based data processing interface (Fig. 1b) of the physical environment system, the view is changed to the Representative cell explorer, and 64 representative cells are created. When each representative cell is created, an inset generation event occurs. At this time, mesh data is generated from each cell volume data through the marching cube algorithm, and the generated mesh data is transmitted to the virtual environment.

**Interaction:** With the left-hand *Focus* interaction (red sphere of Fig. 1c) with hologram insets, the physical environment displays the corresponding cell in the Visualization view. In this way, the user can check the 3D shape and detailed information of the cell at once. With the left-hand *Select* interaction (green sphere of Fig. 1c), the physical environment adds the cell to the selected list which is used as training data for a random forest to classify target cells.

**Connected position:** A connected position of an inset is the coordinate of the cell on the 3D volume rendering in Visualization view. Therefore, the leader line of the inset points to the position of the cell in 3D volume rendering. With the pointing information, users can find cells in a specific region.

## 3 RESULTS

To evaluate the usability of *HoloInset*, we conducted a user study for ZeVis with *HoloInset* system with six visualization researchers (three males and three females between the age of 22–45 years). The study was conducted in a controlled lab environment, and they used a Microsoft HoloLens2 and general desktop environment with a 1980×1080 resolution 27-inch monitor and physical input devices.

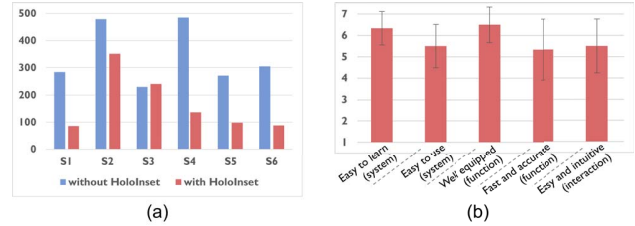


Figure 3: (a): Task completion time for ZeVis system without and with *HoloInset*. (b): Average usability scores for ZeVis with *HoloInset* (1: strongly disagree - 7: strongly agree,  $\pm$  standard deviation).

They received a brief introduction about each visualization system and *HoloInset* and were allowed a short practice time. Once the introduction is over, they were asked to perform a specific analysis task on each system using two different settings, one with only the conventional analysis system and the other with the system combined with *HoloInset*. The task completion time was measured for quantitative performance evaluation.

The participants were asked to select normal cells (objects with successfully generated labels) located in the left brain (cells with an x-coordinate greater than half) from among 64 representative cells generated through the *Representative cell generation* function.

As a result, the participants showed faster task completion time through the proposed system (average of 167 seconds) compared to the conventional system (average of 342 seconds) (Fig. 3a) and rated the usability of the proposed system highly (Fig. 3b).

## 4 CONCLUSION

In this paper, we introduced *HoloInset*, a new visualization method for mixed reality to effectively connect 2D physical and 3D virtual environments for 3D biomedical image analysis. We demonstrated its usability and effectiveness by applying it to the existing visual analytics system and conducting a user study. We believe *HoloInset* has shown the potential of mixed reality for expanding the visualization space of conventional visual analytics systems.

In the future, we intend to develop and provide a high-level programming API of *HoloInset* for easier and wider applications.

## ACKNOWLEDGMENTS

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## REFERENCES

- [1] J. Choi, D. G. C. Hildebrand, J. Moon, T. M. Quan, T. A. Tuan, S. Ko, and W.-K. Jeong. ZeVis: A visual analytics system for exploration of a larval zebrafish brain in serial-section electron microscopy images. *IEEE Access*, 9:78755–78763, 2021. doi: 10.1109/ACCESS.2021.3084066
- [2] B. Hoppenstedt, T. Probst, M. Reichert, W. Schlee, K. Kammerer, M. Spiliopoulou, J. Schobel, M. Winter, A. Felnhofer, O. D. Kothgassner, et al. Applicability of immersive analytics in mixed reality: Usability study. *IEEE Access*, 7:71921–71932, 2019.
- [3] F. Lekschas, M. Behrisch, B. Bach, P. Kerpedjiev, N. Gehlenborg, and H. Pfister. Pattern-driven navigation in 2D multiscale visualizations with scalable insets. *IEEE transactions on visualization and computer graphics*, 26(1):611–621, 2019.
- [4] L. Pavanatto, C. North, D. A. Bowman, C. Badea, and R. Stoakley. Do we still need physical monitors? an evaluation of the usability of ar virtual monitors for productivity work. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, pp. 759–767. IEEE, 2021.
- [5] M. Venkatesan, H. Mohan, J. R. Ryan, C. M. Schürch, G. P. Nolan, D. H. Frakes, and A. F. Coskun. Virtual and augmented reality for biomedical applications. *Cell Reports Medicine*, 2(7):100348, 2021.